

Data-Driven Congestion Management under Uncertainty

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Abstract

Policy for load management in energy networks is highly complex due to intricate relations between key factors, strong autocorrelation, many zero occurrences, and volatile load distribution. In this paper, we provide a data-driven framework that accounts for these challenges to forecast the expected total load of redispatch one day-ahead, one week-ahead and one month-ahead. We propose a two-step model approach that first classifies whether redispatchment occurs and if so, in the second step how large the total load will be. We identify key drivers of the total load, revealing strong dependence on historical load information. Our models include time-varying parameters to maintain flexibility for periodic changes in the impact of drivers. We use a conformal inference technique to build confidence intervals around the forecast of expected total load from our complex models. We find that uncertainty is large for total load predictions across forecasting horizons and provide conservative policy recommendations to address this issue.

Keywords: congestion management, data-driven policy, score-driven filters, conformal inference

JEL Classification: C22, C51, D81

1 Introduction

Reliable energy networks are crucial for the modern society, connecting the suppliers and consumers of energy. With the ongoing energy transitions in many countries, the energy system has to adapt its reliance on fossil fuels onto renewable and alternative forms of energy. Such changes stress the importance of flexibility of energy networks. However, there is a tradeoff between flexibility and costs. With greater flexibility towards changes in demand, the transmission system operators (TSOs) need to change operation load frequently, increasing costs and in the end also increasing service charges for consumers. Balancing this tradeoff between flexibility and feed-in of TSOs is particularly challenging because of congestion. This occurs whenever there is too much energy flowing through the grid and as a result the TSOs have to redispatch more or less energy, depending on their location. Redispatching loads of energy is costly and preemptive redispatch policy is therefore crucial for a reliable energy network.

To illustrate the challenges for load management in energy networks, we present the daily and monthly average of total load in the Tennet DE energy network in Germany from 2020 to 2023 in Figure 1. We observe that throughout many days the average load has large peaks, revealing strong autocorrelation across hours. It highlights that the system was not able to anticipate on demand changes throughout the day. Non-linear, complex relations between the target variable of policy and the predictors may be present in the data, requiring a flexible modelling strategy. Moreover, there is a persistent and seasonal trend visible in the monthly average. This observation stresses the importance of accounting for trends and seasonality, that may even be changing structure over time. Furthermore, the daily average of total load seems to be volatile, switching magnitude of peaks, which reflects large uncertainty about how demand might be in the future.

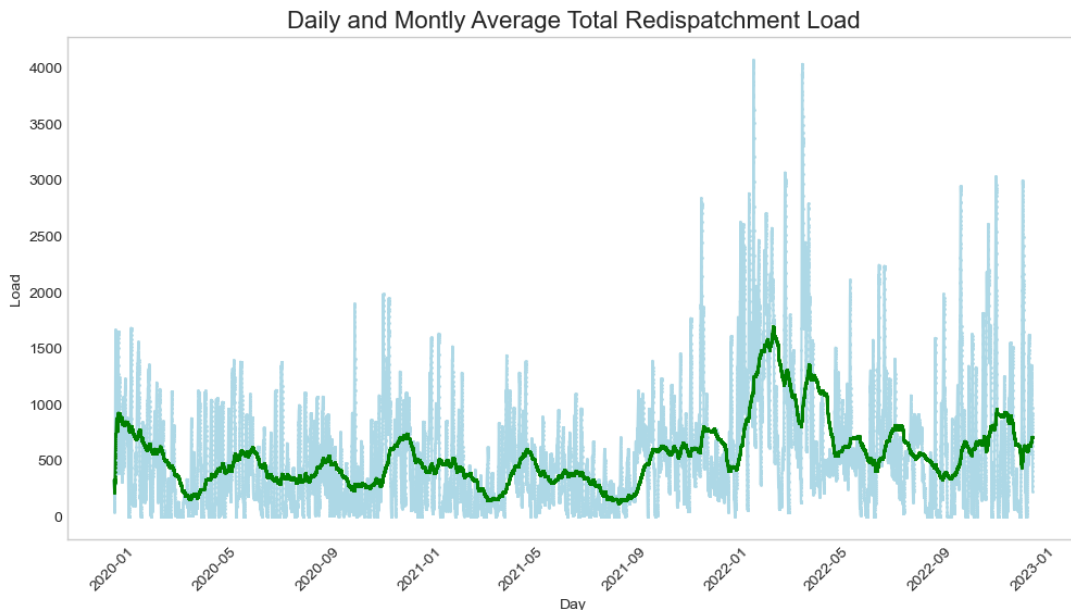


Figure 1: Daily frequency of hours with at least one redispatch.

As illustrated above, the energy environment is highly complex and uncertain. Though uncertain, policy needs to be made to adapt the system to changes in the environment. This is challenging since misjudgements can result in large amounts of irreparable construction costs. For decision-making this posits a challenge: we need to provide data-driven, flexible models to anticipate demand changes, while complex demand dynamics and large black-box AI models introduce uncertainty into decisions. To address these ongoing challenges for congestion management, we aim to answer the following research question:

How can we build preventive congestion policy decisions under uncertainty?

We aim to answer our research question using the following two objectives. First, we aim to build a model that is flexible, robust and interpretable:

- **Flexibility.** The model needs to be flexible toward changing demands of consumers, causing congestion. In different time periods, different factors may be important to predict congestion. Furthermore, the models needs to be able flexible for large amount of predictors while preventing overfitting. Such flexibility may come with a minimal cost of predictive performance of the likelihood of redispatchment.
- **Robust.** The model needs to be robust toward short-term and long-term objectives. Often, policy-makers have multiple objectives, requiring to prevent overfitting of a single horizon. We propose to train separate models for each objective to circumvent this problem.
- **Interpretable.** For policy, it is crucial to know what policies can be constructed on which variables, especially when uncertainty is large. We propose a model that is easy to interpret and provide valid uncertainty bands for a data-driven approach.

Second, for preemptive policy, it is crucial to identify the important drivers impacting congestion and thus the total load of redispatchment. There exists a vast literature about possible important exogenous variables. However, similarly to the factor zoo in finance, there may only be a selection of variables that structurally explains congestion behavior. We aim to identify such structural drivers to build policy.

Related literature. Recent work on congestion forecasting has increasingly focused on combining flexible machine learning models with uncertainty quantification and interpretability. [Zhou and Bialek \(2011\)](#) offer one of the earliest approaches to short-term congestion forecasting, while [Titz et al. \(2024\)](#) apply explainable AI to identify drivers of redispatch in the German grid. Other studies emphasize the need for calibrated uncertainty estimates. For instance, [Gürses-Tran et al. \(2023\)](#) use quantile regression and LSTM models to forecast load distributions, and [Hernandez-Matheus et al. \(2024\)](#) combine probabilistic power flow with ML to assess congestion risk. To reduce computational

complexity, [Liao et al. \(2025\)](#) propose an efficient tree-based method for real-time prediction and control.

Building on this literature, we propose a two-step approach that integrates data-driven techniques such as classification models, score-driven filters and conformal inference to first forecast the probability of congestion and in the second step forecast the magnitude of the necessary load in ever-changing energy environments. This allows us to forecast congestion probabilities, quantify predictive uncertainty, and identify key drivers-all in support of preventive policy design under uncertainty.

Outline. The paper is organized as follows. Section 2 provides details about the TenneT DE network data. Section 3 outlines the methodology for forecasting and identifying important drivers of the expected total load of redispatchment. The results are shown in Section 4. In Section 5 we discuss the role of uncertainty for policy making. Section 6 concludes.

2 Data Description

We use the time series data for the TenneT DE Network, one of the four electricity networks within Germany. The data spans the period January 1, 2020 to July 31, 2023 on an hourly frequency. Importantly, the data records the direction of redispatchment (upward/downward), the number of concurrent redispatches, and the total load of redispatchment. The latter two variables are zero if supply and demand in the network are level, which implies that no congestion occurs. Congestion occurred when the number of concurrent redispatches is positive and hence the total load is positive. Therefore, we define our target variable **expected total load** as a non-negative continuous variable. In the following sections, we investigate the properties of the target variable and the predictor variables, until 31st of December 2022 (corresponding to the train and validation set, see Section 3 for details).

2.1 Target Variable

We aim to forecast the expected total load in any direction, since we aim to construct policy irrespective of direction of redispatchment. Figure 2 below shows the (i) the daily, weekly, and monthly average of total load, (ii) average total load per month in the year, (iii) average total load per day in the week, and (iv) average total load per hour in the day. We observe at least four seasonal patterns. First, we observe strong serial dependence in the total load. For example, the rolling monthly average shows slow up- and down movements, suggesting that once congestion occurs it is likely to occur for a long period after and that the corresponding total load is adjusted slowly. Second, in the top-left figure we observe that the daily average is frequently large and that some days have no load. This observation suggests that within 24 hours, congestion does not change frequently. This is confirmed

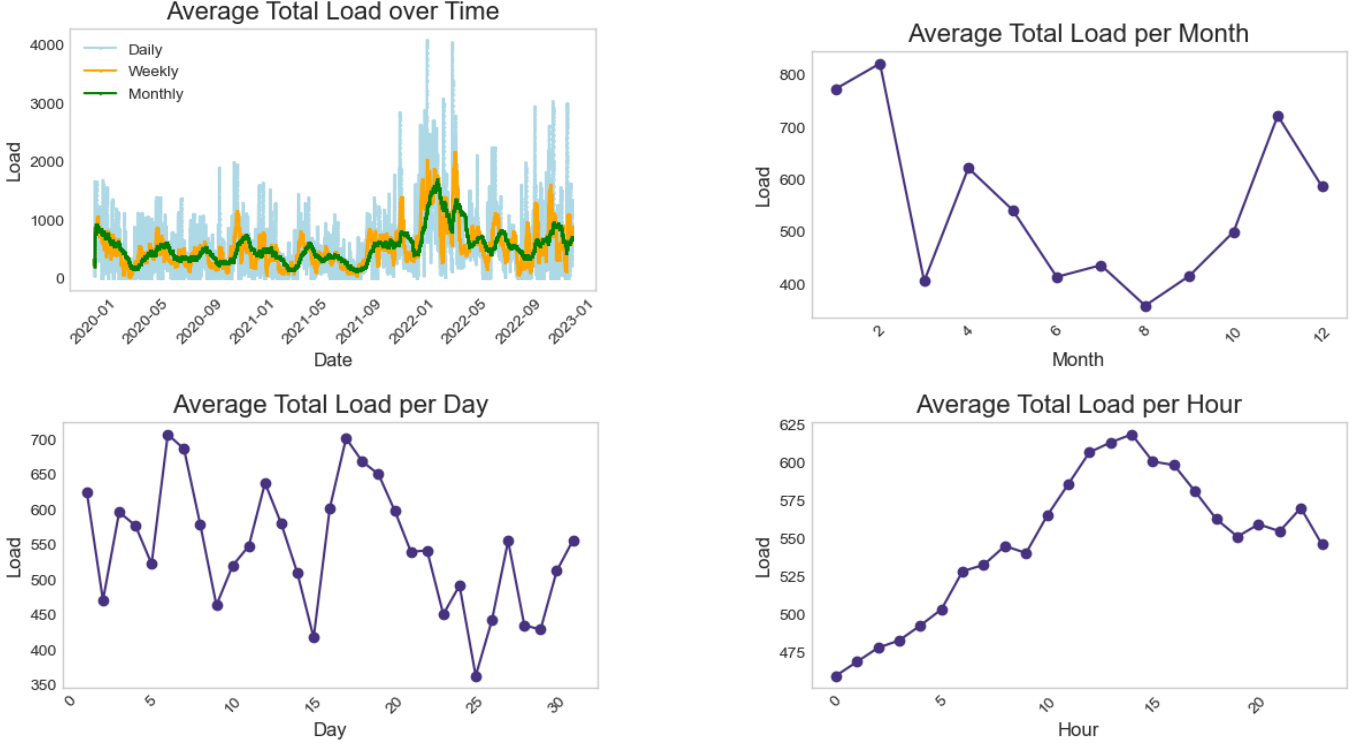


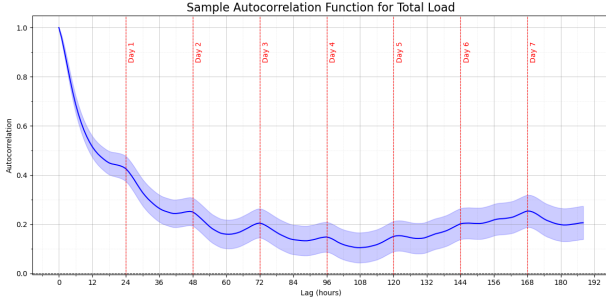
Figure 2: This figure shows the total load data in the following ways: (i) daily, weekly and monthly average of the total load, (ii) average total load per month, (iii) average total load per day in the week, and (iv) average total load per hour in the day.

by an average of 75.9% redispatches in our data. From the first and second observation we know that, in order to develop an accurate forecasting model for both short-term and long-term horizons, the model must account for long trends and the lack of redispatchment switching within a day.

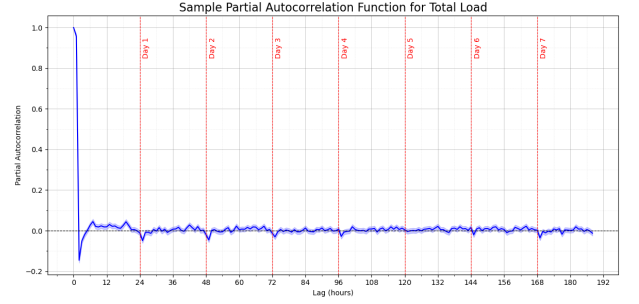
Third, from the total load frequency per month and per hour, we observe additional seasonal patterns. We see that in the winter months (months 1-2, 11-12) a higher load of redispatchment occurs. Energy consumption is higher in the winter due to cold weather, and thus, grid operators need to balance the large demand by more frequent spikes of supply. At the same time, an increase of downward redispatchment may occur due to variable weather. More and stronger winds may generate more energy than in the summer, creating oversupply and hence, more frequent downward redispatchment that contribute to the total load. Similarly, there is less solar energy available in the winter, making alternative sources more important and therefore balancing more difficult. Moreover, we see that during the afternoon (hour 10-17) the average total load of redispatchment is higher. Since consumers are usually inside during these hours (e.g. at work), demand for energy increases.

Fourth, during the days of the month the total load seems relatively stable, with some local spikes and troughs. Since we see in the top-left figure that there exists a slow monthly dependence, this stability suggests that there are also little structural deviations from this trend.

We further examine the serial dependence by computing the sample (partial) autocorrelation function for total load upto a lag of 188 hours (slightly more than one week) with standard chi-square



((a)) ACF



((b)) PACF

Figure 3: This figure shows the sample (partial) autocorrelation function for the total load data.

critical values, see Figure 3. The sample autocorrelation function illustrates large, significant lags in congestion for all lags. However, the partial autocorrelation function shows that after one lag the correlation is very close to zero and mostly insignificant. To capture the trending pattern visible in the autocorrelation function and as already highlighted above Figure 2, we extend the dataset with the following variables: the lagged value of total load and the average total load in the past 24 hours (day), in the past 168 hours (week) and in the past 720 hours (month), as far as available. Naturally, similar seasonal patterns occur for the binary indicator of redispatchment, so we extend the dataset by equivalently defined historical averages.

2.2 Predictor Variables

The dataset comprises 76 variables that are potential factors impacting the total load of redispatchment. Importantly, from an initial analysis, we observe that the financial prices are non-stationary with long trends. Therefore, we compute first-differences of all prices available and work with these instead. Intuitively, a positive first-difference implies that financial investors expect that the demand for the underlying asset such as energy is growing. We divide the variables into three categories: weather factors, electricity factors and market factors. We visualize the time series for a selection of variables in Figure 4. The seasonality of the wind data and the electricity forecast seem to follow each other. Similarly, the consumption and the day-ahead German electricity price follow each other closely.

For policy, it is important to identify drivers of the total load. As an initial step, we visualize the correlation matrix across all variables available including the additional variables described in the previous subsection, see the left panel in Figure 5. The blocks of correlation correspond to variables that measure similar objects such as historical averages and closely related prices of electricity patterns, as also seen in Figure 4. In the right panel we zoom-in on a selection of variables with large correlation with total load in Figure 5). We exclude redispatchment variables from this matrix since they are trivially correlated with total load. We see that the lagged total load has the largest correlation with the total load, conforming to the strong serial dependence in total load. At first glance, the exogenous

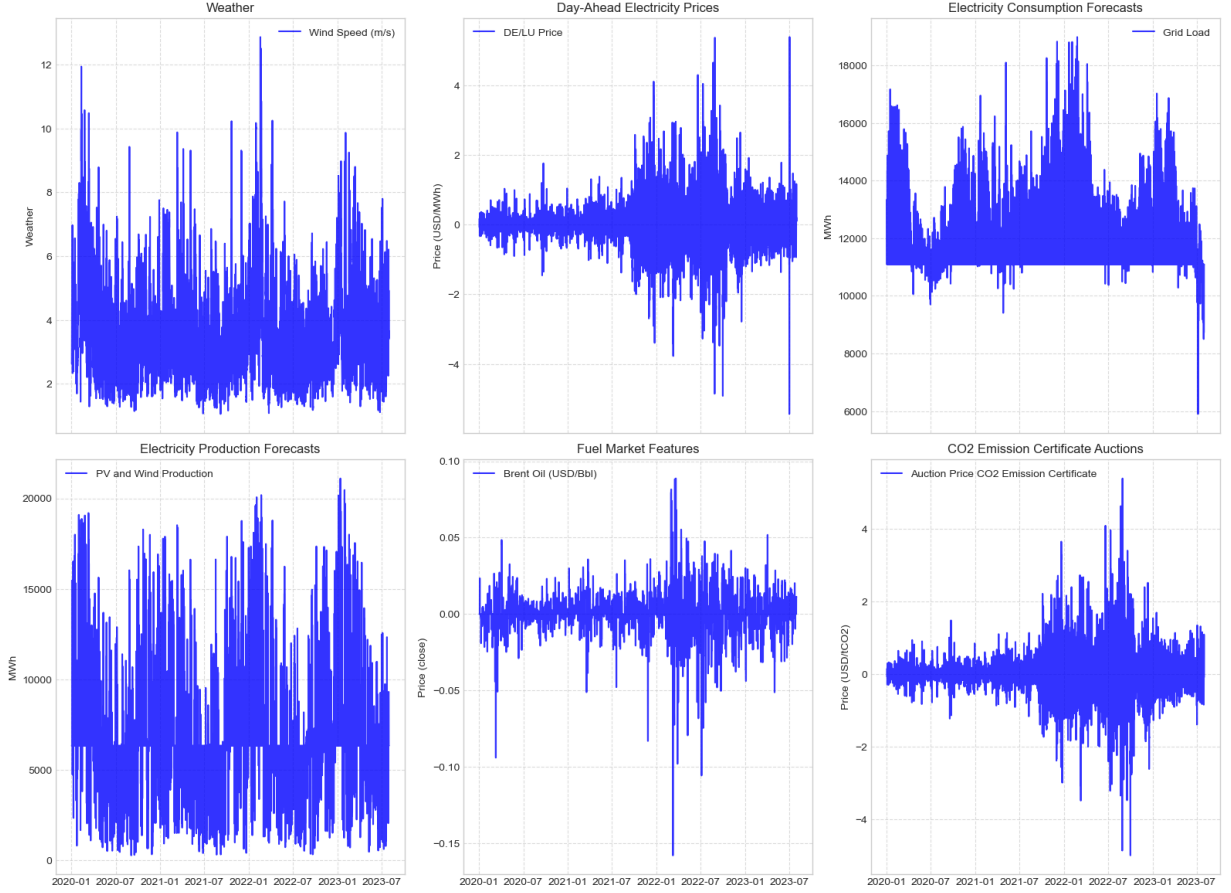


Figure 4: This figure shows a selection of predictor variables. For each data category, we show at most two variables.

variables such as wind speed and temperature show little correlation with the total load. Similarly, seasonal indicators for weekend, day, month, and hour of the day, seem to have little relation with total load and are excluded from the matrix. We conjecture for our model that any seasonal effects may be sufficiently captured by lagged values. Therefore, a benchmark predictive model for total load may be running average across the past 24 hours, see Section 3 for details.

The correlation shown in Figure 5 are static statistics, but the correlation between a factor and the total load may be time-varying. Figure 6 depicts the correlation over a 188 days window between congestion and the five variables that are key factors from the literature. We see that the correlation seems to be time-varying and exogenous variables may gain importance in some episodes. This suggests that our data-driven forecasting model should incorporate a possibly time-varying effect, for example, making use of moving-window training approach and time-varying parameters. Moreover, policy should be flexible to anticipate on an instable environment, implying that timely monitoring of factors is crucial. We elaborate on our policy suggestions in Section 5.

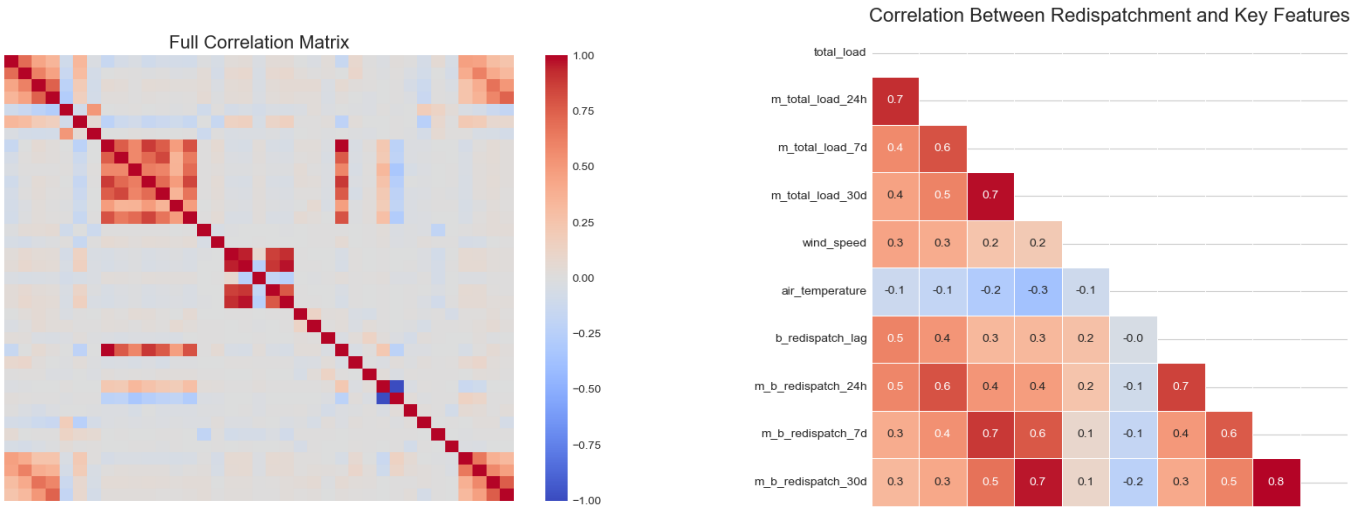


Figure 5: This figure shows the correlation matrix for the target variable and predictor variable. The right figure shows the correlation matrix for a selection of variables with large correlation with total load.

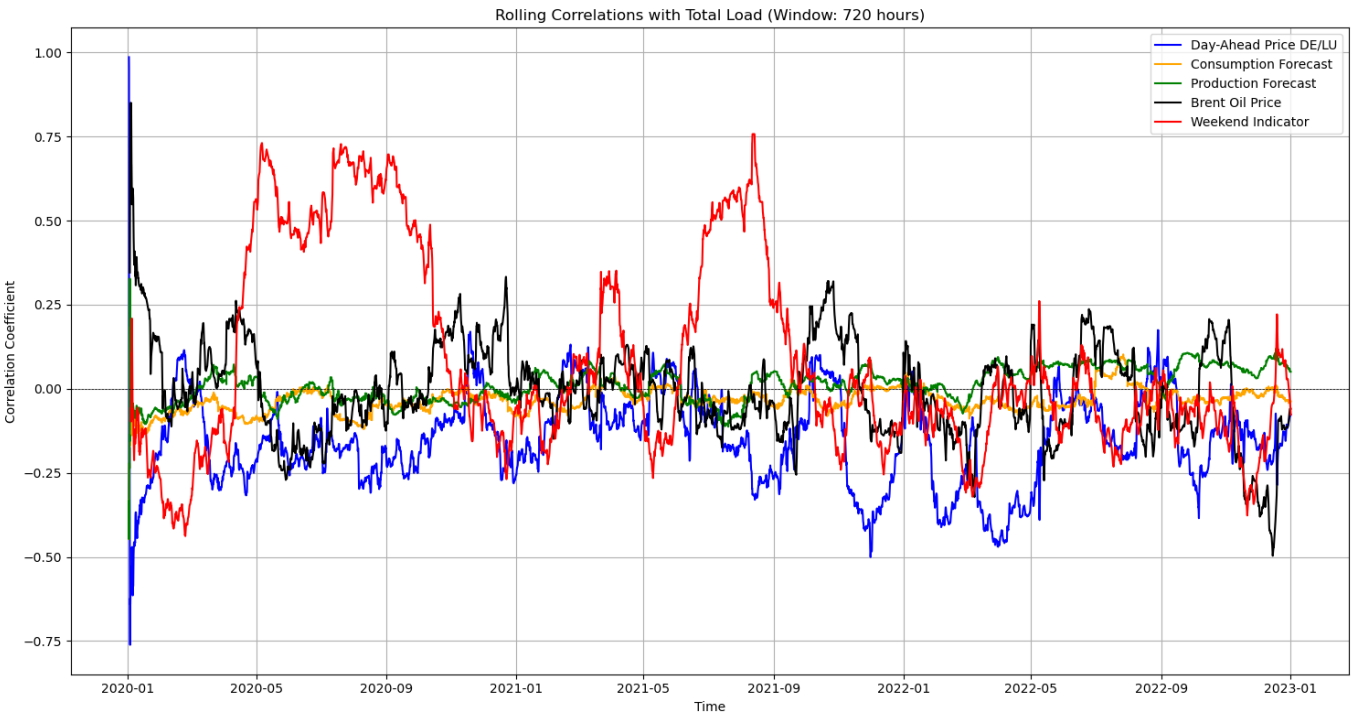


Figure 6: This figure shows the time-varying correlation for a selection of key factors with congestion. The upper panel show the correlation for upward direction of congestion. The lower panel show the correlation for downward direction of congestion.

3 Methodology

In this section, we describe the methods used to investigate our two main objectives: (i) to accurately forecast the total redispatch load one day, one week, and one month ahead on an hourly basis, and (ii) to provide relevant policy recommendations based on these forecasts.

For the first objective, we propose a two-step forecasting approach. In the first step, we estimate the probability of a congestion event occurring at each forecast horizon. In the second step, we predict the total redispatch load conditional on congestion being present. The product of these two components yields the unconditional expected redispatch load. Our modeling strategy is illustrated in Figure 7. Lastly, we describe how the point forecasts are evaluated.

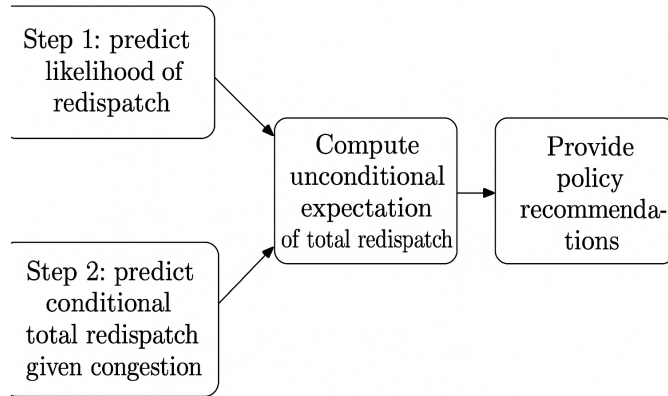


Figure 7: Overview of the two-step forecasting framework. First, we estimate the probability of congestion. Then, conditional on congestion, we predict the expected redispatch. The product yields the unconditional expectation, which informs our policy recommendations. Figure is generated by ChatGPT.

3.1 Modeling Strategy

We aim to accurately predict the target variable y_t , which is the sum of the total load in both up and downward direction at hour t . We aim to forecast this variable one-day, one-week and one-month ahead (for a month we set fixed at 30 days). Mathematically, we consider forecast horizons $h \in \{1 \times 24, 7 \times 24, 30 \times\} = \{24, 168, 720\}$ for which we make a prediction at time t . Therefore, we write our prediction in vector form, as follows:

$$\hat{\mathbf{y}}_t = [\hat{y}_{t+24|t}, \quad \hat{y}_{t+168|t}, \quad \hat{y}_{t+720|t}]^\top.$$

where $\hat{y}_{t+h|t}$ is the expected value of the variable h -hours ahead:

$$\hat{y}_{t+h|t} := \mathbb{E}[y_{t+h} | \mathcal{F}_t] := \mathbb{E}_t[y_{t+h}],$$

where $\mathcal{F}_t$ is the information available at hour t . We model each forecast horizon h separately, resulting in three total models. As the target variable is often zero, i.e. oftentimes there is no redispatch at all, we adopt a two-step approach. To do this, we first define a latent binary variable $z_t \in \{0, 1\}$ as follows:

$$z_t = \begin{cases} 1, & \text{if } y_t > 0 \\ 0, & \text{if } y_t = 0 \end{cases}$$

Note that the case $y_t < 0$ is not considered as y_t is non-negative. We then apply the tower property of conditional expectations:

$$\hat{y}_{t|t+h} = \mathbb{E}_t[y_{t+h}] \quad (3.1)$$

$$= \mathbb{E}_t[y_{t+h}|z_{t+h} = 1]\mathbb{P}[z_{t+h} = 1] + \mathbb{E}_t[y_{t+h}|z_{t+h} = 0]\mathbb{P}[z_{t+h} = 0] \quad (3.2)$$

$$= \mathbb{E}_t[y_{t+h}|z_{t+h} = 1]\mathbb{P}[z_{t+h} = 1], \quad (3.3)$$

where the second term of Equation 3.2 is equal to zero (by definition). We then model these two terms separately. Specifically, we estimate $\mathbb{P}[z_{t+h} = 1]$ by a logistic regression, which is explained in Section 3.2.1. We estimate $\mathbb{E}_t[y_{t+h}|z_{t+h} = 1]$ by considering a linear regression where we use only the positive values of the target variable. We consider both a static and time-varying approach, which are discussed in Section 3.3.1 and 3.3.2, respectively.

3.2 Step 1: Forecasting the Probability of Redispatch

3.2.1 Logistic Regression

We consider the estimation second term of Equation 3.3 as a binary classification task. In that regard, we can estimate it by using a logistic regression, which is an attractive choice due to its interpretability, and is widely used in both econometrics and forecasting (Carrasco, 2001; Asif and McHale, 2016). We define $\hat{p}_{t+h|t} := \mathbb{P}_t[z_{t+h} = 1]$ as the probability that there is at least one redispatch occurring. Mathematically, the logistic regression models this probability as follows, where we omit the subscript h in the parameters for conciseness:

$$\hat{p}_{t+h|t} = \mathbb{P}(z_{t+h} = 1 \mid \mathbf{x}_t) = \frac{\exp(\beta_0^* + \mathbf{x}_t^* \boldsymbol{\beta}^*)}{1 + \exp(\beta_0^* + \mathbf{x}_t^* \boldsymbol{\beta}^*)}, \quad (3.4)$$

where $\mathbf{x}_t^*$ is the vector of covariates (features) at time t , and $\boldsymbol{\beta}^*$ and β_0^* are the coefficients for the logistic regression model. To account for the curse of dimensionality, we use the 5 variables have the largest correlation with the target variable, following the recommendation of Koprinska et al. (2015). This results in the hourly lag of z_t , the daily, weekly, and monthly historical averages, and wind speed.

One attractive feature of the logistic regression is that it enables to interpret the coefficients. In our research, we evaluate the magnitude of the coefficients, and to test for significance we use the Wald test, and for details, we refer to [Wald \(1943\)](#).

To account for the imbalanced class distribution in the redispatch dataset, we incorporate weighted likelihoods in the estimation procedure, assigning higher weights (inversely to its proportion) to the minority class (i.e., periods of non-congestion). This guides the model to better capture the characteristics that are relevant for that minority class.

3.3 Step 2: Forecasting Conditional Redispatch Load

3.3.1 Linear Regression

For the estimation of the first term in Equation 3.3, we use a linear regression. As y_t conditioned on $z_t = 1$ is positive by definition, we use $\log(y_t)$ as the dependent variable (we omit conditioning on z for brevity). Then, the linear regression equation is given as follows:

$$\log(y_{t+h}) = \beta_0 + \mathbf{x}_t^\top \boldsymbol{\beta} + \varepsilon_t, \quad (3.5)$$

where $\mathbf{x}_t$ is the vector of covariates for the linear regression model, and β_0 and $\boldsymbol{\beta}$ are the coefficients for the linear regression. For the vector of covariates $\mathbf{x}_t$, we use the rule of thumb of [Krehbiel \(2004\)](#) for variable selection, resulting in 16 covariates with the highest correlations to the target variable. As by Jensen's inequality $\mathbb{E}[\log y_t] \geq \log \mathbb{E}[y_t]$, we can not simply take the exponential to obtain the prediction of y_t . Therefore, we assume $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$, which results in $\log(y_t)$ being log-normally distributed. We can then take $\mathbb{E}[y_t] = \exp(\mathbb{E}[\log y_t] + \frac{1}{2}\sigma^2)$ to obtain an estimate of the target variable. For estimation, we use Ordinary Least Squared (OLS).

3.3.2 Score-Driven Linear Regression

To allow for the time variation in the effects of the covariates on predictions suggested by our data analysis, we use of a time-varying parameter model. Such models are typically classified into parameter-driven and observation-driven approaches; see ([Cox et al., 1981](#)) for a discussion of this distinction. For this setting, we deem an observation-driven model more appropriate due to its fast computation time, flexibility, and good empirical performance ([Koopman et al., 2016](#)). Specifically, we adopt a score-driven approach ([Harvey, 2013](#); [Creal et al., 2011](#)), which has proven to be successful across a wide range of applications (see www.gasmodel.com for references).

We model the total redispatch load y_t using a log-normal regression with time-varying coefficients

β_t . The log-normal specification ensures that the predicted outcome remains strictly positive and accommodates skewed distributions. We assume

$$\log y_t = \beta_{0,t} + \mathbf{x}_t^\top \boldsymbol{\beta}_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2),$$

so that y_t is conditionally log-normal given the covariates $\mathbf{x}_t$, and β_0 and $\boldsymbol{\beta}$ are the coefficients. Each coefficient $\beta_{j,t}$ moves dynamically according to a GAS(1,1) update equation:

$$\beta_{j,t+1} = \omega_j + \alpha_j s_{j,t} + \beta_j \beta_{j,t},$$

where $s_{j,t}$ denotes the scaled score of the log-likelihood with respect to $\beta_{j,t}$:

$$s_{j,t} = \left(\frac{1}{\sigma^2} \right)^{-1} \cdot \left(\frac{\log y_t - \mathbf{x}_t^\top \boldsymbol{\beta}_t}{\sigma^2} \right) \cdot x_{j,t} = (\log y_t - \mathbf{x}_t^\top \boldsymbol{\beta}_t) \cdot x_{j,t}.$$

We compute the score by taking the gradient of the log-likelihood from the assumed log-normal model and scale it by the inverse Fisher information as suggested in [Creal et al. \(2013\)](#)

We estimate all the static parameters in our model, the drift ω_j , the score loading α_j , and the persistence parameter β_j for each feature j , and the log-variance $\log \sigma^2$ by maximum likelihood which is both standard in the literature and practically efficient ([Artemova et al., 2022a,b](#)) following [Blasques et al. \(2022\)](#). This framework provides a flexible specification, yet allows straightforward interpretation of the (time-varying) influence of each of the covariates.

More sophisticated extensions of the time-varying parameter model we use are available, including robust variants of the score-driven filter ([Harvey and Luati, 2014](#)) and simulation-based approaches such as the (auxiliary) particle filter ([Pitt and Shephard, 1999](#)). However, we believe our chosen specification strikes a good balance between empirical performance and time efficiency.

3.3.3 Baseline: Random Walk

As a simple benchmark that is often used in the forecasting literature, we consider a random walk model to predict the total redispach load. This model assumes that the best predictor of any future value is the most recently observed value, and is widely used due to its simplicity and strong empirical performance, which holds especially in settings where the target variable exhibits high autocorrelation (see ([Hyndman and Athanasopoulos, 2018](#))). Formally, for each forecast horizon $h \in \{1, \dots, H\}$, the h -step ahead forecast at time t is given by

$$\hat{y}_{t+h|t} = y_t,$$

where y_t denotes the observed total redispatch load at time t .

3.4 Performance Evaluation

3.4.1 Root Mean Square Error

To evaluate the accuracy of our hourly predictions across the three forecast horizons, we use the Root Mean Squared Error (RMSE), which measures the average difference of the prediction error by computing the square root of the mean squared differences between the observed redispatch load and its forecast. The RMSE is well suited to our setting, as it penalizes larger errors more heavily, which is an important property given the operational consequences of strongly under- or overestimating redispatch volumes.

3.4.2 Uncertainty Quantification

To highlight the uncertainty surrounding the point forecasts we use Conformal Inference (CI) as introduced by (Vovk et al., 2005). This model-agnostic framework constructs prediction intervals especially useful for complex, non-linear models with formal coverage guarantees. Classical CI methods require minimal distributional assumptions, but rely on the exchangeability of observations, which is violated in time series settings due to temporal dependence and distributional shifts. Recent adaptations of standard CI methods address this issue, for instance, Adaptive Conformal Inference (ACI) (Gibbs and Candes, 2021), Ensemble Prediction Interval (EnbPI) (Xu and Xie, 2023), and Error-Quantified Conformal Inference (ECI) (Wu et al., 2025). We adopt ECI, as it is the only method among these that provides *finite-sample* guarantees, making it applicable to our binary dataset. Additionally, ECI offers sharper and more stable prediction intervals by incorporating continuous error feedback (Wu et al., 2025).

4 Results

This section presents the main findings of our two-step modeling approach. We train our models based on a training period from 1 January 2020 to 31 December 2022, and evaluate our approach on a training period spanning 1 January 2023 to 31 July 2023. Our approach does not entail hyperparameters, which allows to use a larger training set for training since a validation set is not needed. Any static parameters in our model such as (ω, α, β) for the score-driven filter are trained once on the training set and are not updated sequentially. Such a procedure provides fast forecasts, which makes fast preemptive action possible. We use the performance metrics described in Section 3 to evaluate the forecasting performance of the first and second step.

4.1 Step 1: Performance of Likelihood Forecasting

We first compare the out-of-sample performance of the logistic regression and the historical average on the test set. The accuracy and cross-entropy results for one day-ahead, one week-ahead, and one month-ahead are given in Table 1.¹ We see that the simple historical average makes reasonable accurate one-hour ahead forecasts, which we can attribute to the high serial dependence in the binary redispatch data. We observe that the logistic regression achieve higher one day-ahead forecasts than the historical average, but this difference diminishes as the horizon increases. These results indicate the following. The results suggest that a parameter-free historical average is already capable of capturing the main predictive signals present in the redispatchment data for large horizons, but on a short horizon predictor variables such as hourly lag and daily lag in the logistic regression provide additional explanatory power, see Table 2. As illustrated in Figure 3, the serial dependence is decreasing and hence, recent historical averages becomes less informative for predicting further ahead redispatch events.

	Historical Average		Logistic Regression	
	Acc.	CE	Acc.	CE
Day	0.433	4.02	0.700	0.600
Week	0.419	4.99	0.655	0.641
Month	0.411	6.47	0.603	0.623

Table 1: This table shows the accuracy and cross-entropy scores for the historical average frequency and our logistic regression model, across daily, weekly and monthly horizons

Variable	1-day ahead		1-week ahead		1-month ahead	
	Coeff.	<i>p</i> -value	Coeff.	<i>p</i> -value	Coeff.	<i>p</i> -value
intercept	-2.16	0.00	-1.44	0.00	-0.364	0.00
Hourly lag	1.23	0.00	0.360	0.00	0.0157	0.71
Daily lag	-0.144	0.024	0.592	0.006	-0.161	0.008
Weekly lag	1.20	0.00	0.460	0.00	1.81	0.00
Monthly lag	2.15	0.00	2.46	0.00	0.435	0.0009
Wind speed	0.058	0.001	-0.041	0.002	0.0063	0.55

Table 2: This table shows the estimated coefficients for the logistic regression and their significance.

Moreover, to ensure stability of our two-step approach, it is crucial to identify the periods in 2023 where our logistic regression model is performing well and when it struggled. Figure 8 shows the cross-entropy loss for logistic regression over a rolling window of one week, across horizons. We observe that the cross-entropy remains relatively stable, with some local upward spikes in March 2023 and June 2023. We conjecture that these spikes are related to season changes, since both months are close changing seasons. Overall, our results suggests that the choice for logistic regression is robust.

¹In a previous edition of this paper (Case A), we investigated the balance of our forecasts. The confusion matrix in that version shows that our forecasts are balanced and not over-predicting the majority class, which is one. We omitted the matrix for brevity reasons.

We leave it for further research to construct more complex models that can capture the structural error from the logistic regression but achieves similar stability.

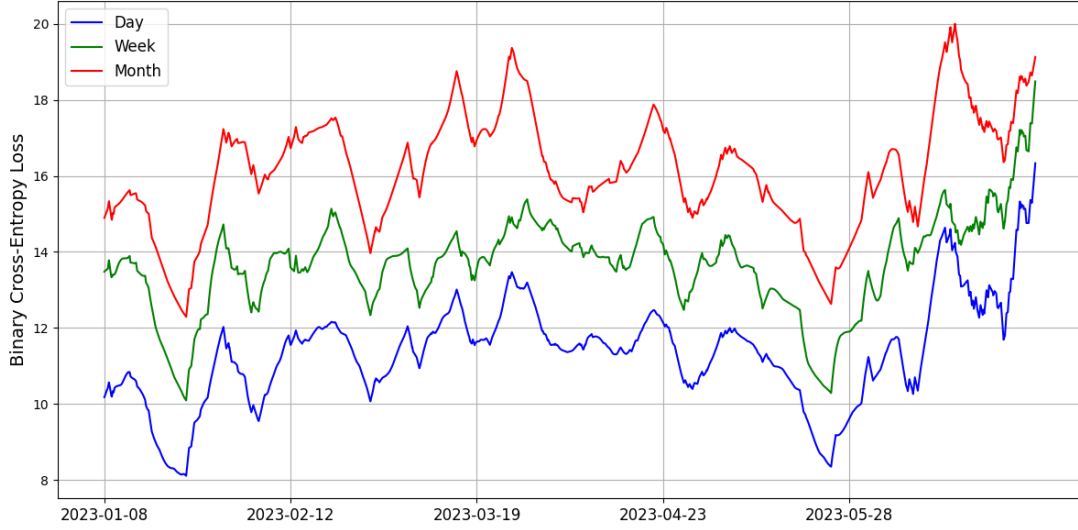


Figure 8: This figure shows the cross-entropy loss of our congestion likelihood forecast across three horizons.

Relying on point forecasts alone can be tricky, especially in settings where decisions involve potentially high costs, as with redispatching. It is therefore important to understand how uncertain our predictions are before using them for policy recommendations. Figure 9 shows the 24 hour ahead redispatch probability forecasts from the logistic regression model with associated 95% prediction intervals from error-corrected conformal inference. The prediction intervals vary in width over time, with some short periods having CI intervals spanning the entire $[0, 1]$ range. Notice that periods with high forecasted congestion probabilities are (still) surrounded by considerable uncertainty. Hence, when making policy decisions, it is important to take this large uncertainty into account as it is costly to redispatch. We refer to Section 5 for a more detailed discussion.

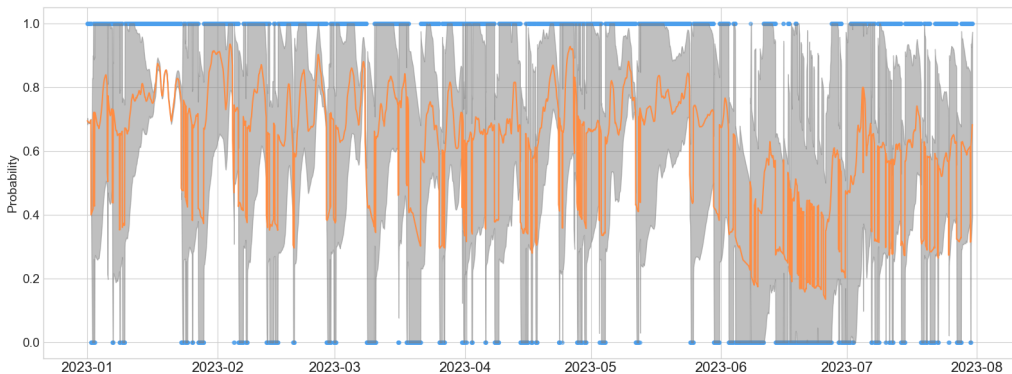


Figure 9: One-day ahead redispatch probabilities predicted using logistic regression (orange), with 95% error-calibrated conformal inference intervals (grey), and the actual redispatch outcomes (blue).

4.2 Step 2: Performance of Total Load Forecasting

Our main goal is forecasting the expected total load. In step two of our approach, the estimated probabilities are used together with positive predictions of total load to compute the final estimate of the expected total load, as outlined in Section 3. The performance of our out-of-sample forecasts are illustrated in Figure 10 over time for one-day-ahead and one-week-ahead predictions. Due to space constraints, we omit the figure for the one-month-ahead forecasting horizons, but our conclusions (below) remain the same.

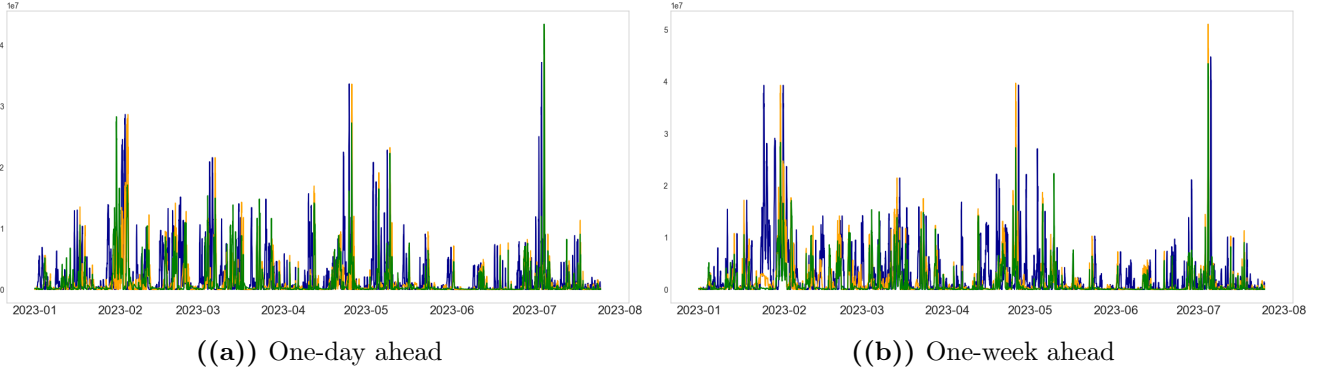


Figure 10: Out-of-sample root mean squared prediction error (RMSE) for the one-day and one-week ahead forecasts from the random walk (dark blue), linear regression (green), and score-driven (orange) models.

The score-driven model almost always provides the lowest RMSE across both forecasting horizons, which confirms our hypothesis that allowing for time-varying covariate effects (i.e., a more flexible model) is appropriate in this setting to improve forecasting accuracy. Moreover, both the random walk and static linear regression models show a sharp increase in RMSE around July 2023, which highlights their inability to adapt to structural shifts in the data. Overall, as expected, the baseline random walk model performs worst across both horizons.

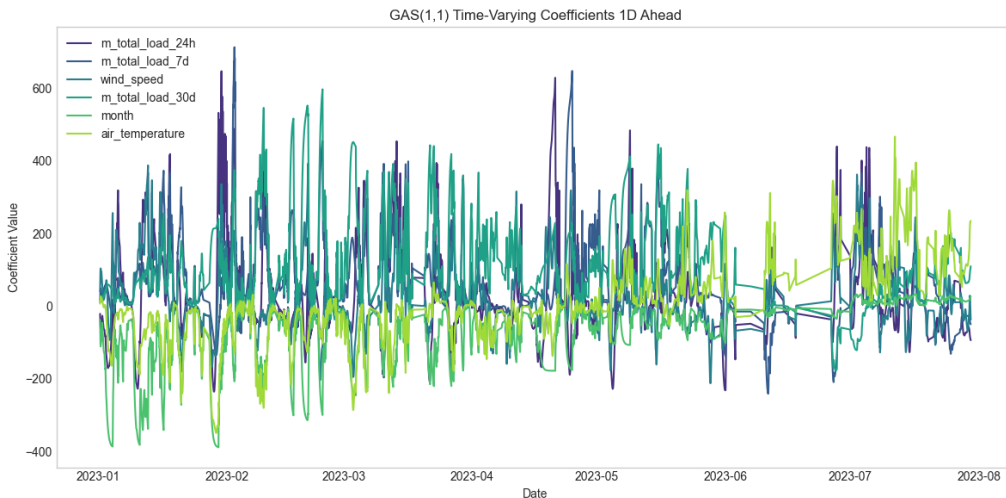


Figure 11: One-day ahead redispatch probabilities predicted using logistic regression (orange), with 95% error-calibrated conformal inference intervals (grey), and the actual redispatch outcomes (blue).

To investigate whether the (implicit) assumption that the covariate coefficients are constant over

time and there is no need for a more flexible model is valid or not, we plot the estimated (i.e., filtered) time-varying coefficients for the one-day-ahead congestion forecasts using our score-driven linear regression model (Figure 11). We see that large variations across all features, with the coefficients frequently switching in direction and clustering together during certain periods. This strongly suggests that the assumption of constant coefficients is not appropriate and supports the need for a flexible, time-varying specification. Furthermore, the coefficient on air temperature is almost always negative, which suggests that higher temperatures are associated with lower congestion risk. In contrast, the wind speed coefficient shows large swings in magnitude, especially early in the year 2023, which is as expected as this is likely to be seasonally. We note that all the coefficients during June and July are less volatile.

Moreover, we compute a single-execution forecast for 2023 using the score-driven regression model. We do one day-ahead predictions. The coefficients follow the recursion update, where the scores are set to zero. We use the predictor variables at December 31, 2022, for each step. In Figure 12 we see that our forecast foresees spikes in February and May, which indeed occurred in hindsight. Furthermore, the single-execution forecast from the score-driven model predicts low values of total load at the end of the test period. This happens because the score values are zero, such that the score coefficients will follow an exponential decay toward a low unconditional mean. Hence, the forecast at the end of 2023 will be low and relatively stable for the single-execution forecast.

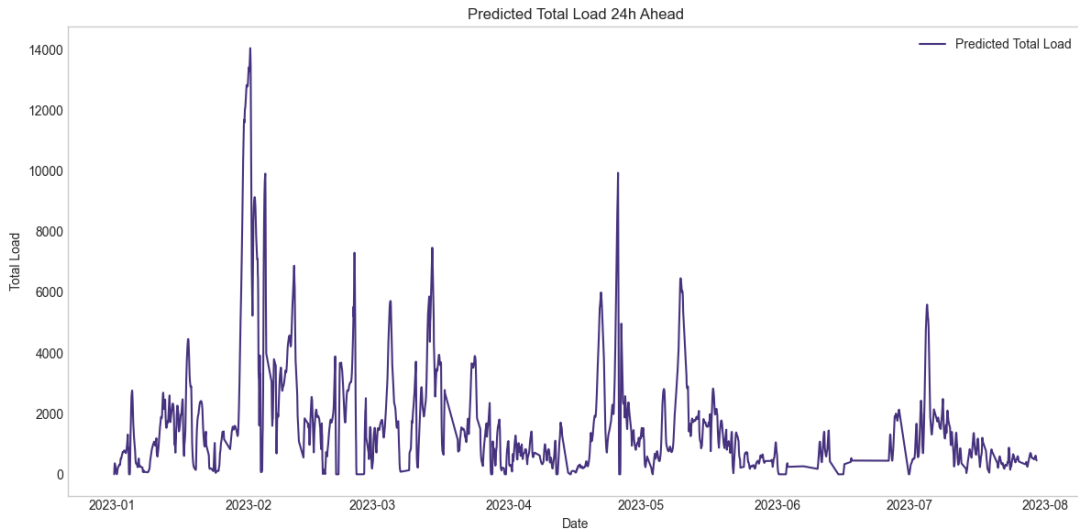


Figure 12: This figure shows the forecasted total load path for 2023 using the score-driven regression model.

5 Policy Recommendations

Our findings show that accurate and robust forecasting of expected total load of redispatchment is possible using a two-step approach with standard logistic regression and linear regression models.

Important factors in the first step are recent historical averages. As argued by [Billault-Chaumartin et al. \(2020\)](#), this importance is explained by the slow operation mode of power plants. The demand load for energy is slowly moving and the TSOs anticipate slowly towards these trends. Reserve plants can mitigate such slow anticipation by capturing the anticipated demand and use the TSOs for flexible demand [Billault-Chaumartin et al. \(2020\)](#).

We have seen that the predicted likelihoods from a data-driven framework like ours come with large uncertainty, even for a simple logistic regression. Since this uncertainty is in our first-step where a binary decision (load management yes or no) is assessed, it serves as a warning for policy-making and hence we recommend conservative policy-making. An example of a conservative policy is nudging/encouraging consumers to reduce demand during peak periods. The cost of such a plan is much smaller than opening a plant and can be repaired/canceled. Similar policies have been implemented for train networks, where such encouragement is preferred over opening more tracks ([Eriksson et al., 2023](#)).

Using large, black-box models for the second step enhances the uncertainty for policy, as many parameters have to be estimated with volatile data. Our findings show that flexible models with time-varying parameters instead of large AI models provide a good solution. Additionally, the paths of the parameters are interpretable and are easy to compute, which can be used to present to stakeholders and further stress testing.

Moreover, the uncertainty suggests that the data and models alone are not sufficiently informative to conduct actions. To build narrower bands and therefore more confidence about our policy, we need to improve the data quality. For example, we can try to improve the data-sharing capabilities between market participants, grid operators, and consumers, so that everyone has access to real-time grid conditions and can better anticipate or react to situations that might lead to congestion. A potential pathway is to construct accurate nowcasts using recent methods such as Bellman filtering ([Lange, 2024](#)).

6 Conclusion

In this paper, we explore the challenges and strategies for load management in energy networks, specifically focusing on preemptive policy decisions under uncertainty. To this end, we develop several models that focus on flexibility, robustness, and interpretability, aiming to meet the need for forecasting the expected total load of redispatchment. Specifically, we propose an interpretable two-step approach where the first step provides probabilities of the need of load management, and in the second step we compute the expected load management given that load management is necessary. We use flexible time-varying parameters and conformal inference to assess changes in relations and

uncertainty for policy.

Our findings indicate that logistic regression outperforms simple averages, and captures the main predictive signals for smaller horizons. Coefficient estimates suggests that the time series is highly persistent, and that lags have more predictive power than covariates. Additionally, we find that the predictions become increasingly uncertain for longer forecast horizons, leading us to recommend a more cautious approach to redispatching, given the challenges in long-term forecasting.

Further research could focus on incorporating uncertainty measures in forecasting total load in our second stage. For example, volatilities in electricity and emission markets could provide insights could help gain further insights on future congestion and total load of redispatchment. Furthermore, we did not include spatial effects in our approach. Alternative data such as satellite weather and emission data could provide earlier insights on changes in energy demand.

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